

Jotheesh Reddy Kummathi

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EXPERIENCE

AI Vision Systems Intern

July 2026 – Present

The Nail Lab, Inc.

New York, NY

- Building an embedded computer vision system on NVIDIA Jetson Orin Nano for an overhead work lamp: real-time detection, segmentation, and tracking on-device in C++ and Python.
- Built a GStreamer pipeline ingesting dual IMX219 sensors over MIPI CSI-2 with hardware-accelerated encode; synchronized capture keeps tracking consistent across overlapping views.

Graduate Research Assistant

Jan. 2025 – June 2026

Agile Robotics and Perception Lab (ARPL) – New York University

New York, NY

- Deployed and calibrated a multi-modal sensor suite (LiDAR, depth cameras, precision GPS) for state estimation in GPS-denied environments; contributed to 3D Gaussian Splatting perception research.
- Brought up PX4 stacks on Jetson Orin companions, root-caused ESC and flight-controller faults reproducible only on hardware, and built HITL/SITL CI/CD pipelines with telemetry logging.
- Safety pilot for UAV flight tests; operated Clearpath UGV and Boston Dynamics Spot platforms in field experiments.

Head Teaching Assistant

Jan. 2025 – May 2026

New York University – ECE-GY 6483: Real-Time Embedded Systems

New York, NY

- Led recitations for 700+ graduate students over three semesters on C, Embedded C, DSP, Mbed OS, and RTOS.

Junior Embedded Engineer

July 2023 – July 2024

Magnibot Technology Solutions Pvt. Ltd.

Bengaluru, India

- Developed and optimized firmware for IoT and embedded products in C/C++ and FreeRTOS, targeting domestic and industrial deployments.
- Designed, debugged, and integrated embedded solutions across UART, SPI, I2C, and CAN; maintained CI/CD pipelines automating build validation and regression testing.

PROJECTS

Slip-Aware MPPI Navigation on Martian Terrain | *Isaac Sim, ROS2, UKF* | [\[GitHub\]](#)

Sep. – Dec. 2025

- Built an autonomous nav stack for a Clearpath Warthog skid-steer platform: slip-aware kinematics, GPS-free UKF localization, MPPI (1500 samples at 10 Hz), 3D LiDAR, and RTAB-Map SLAM.
- Characterized a $2.2\times$ wheel-slip effect the nominal model missed (track width 1.38 m $\rightarrow$ 3.0 m) and folded it into the controller cost to recover tracking on low-friction regolith.

RL Quadruped Locomotion on Unitree Go2 | *Isaac Lab, PPO, Sim-to-Real* | [\[GitHub\]](#)

Sep. – Dec. 2025

- Trained a PPO walking policy in Isaac Lab with multi-term reward shaping (foot clearance, contact forces, torque regularization, Raibert heuristic), reaching ~ 48 linear and ~ 24 angular velocity tracking reward over 6 runs.
- Implemented torque-level PD control with gait-phase tracking and action smoothing; randomized actuator friction and 160-point terrain height scans for sim-to-real transfer.

Person Following on Boston Dynamics Spot | *YOLOv8, ZED 2i, Visual Servoing* | [\[GitHub\]](#)

Jan. 2026

- Deployed real-time visual servoing on physical Spot hardware (YOLOv8 + ZED 2i stereo): 3-axis proportional controller at 10 Hz with deadband filtering, velocity ramping, and a tracking/search/safety state machine.

TECHNICAL SKILLS

Languages: C++, Python, C, Embedded C, MATLAB, Bash/Shell

Robotics & Controls: ROS2, PX4 Autopilot, MAVLink, Motion Planning (A*), Control (PID, MPC, MPPI)

Perception & Estimation: SLAM (RTAB-Map), Kalman Filters (EKF/UKF), Sensor Fusion, LiDAR, IMU, Stereo

ML & Vision: PyTorch, OpenCV, TensorRT, JAX, YOLOv8, 3D Gaussian Splatting, Reinforcement Learning (PPO)

Simulation & Tools: Isaac Sim/Lab, Gazebo, MuJoCo, HITL/SITL, Docker, Git, CI/CD, Linux, GStreamer

Embedded & Hardware: Jetson Orin, ARM Cortex-M, FreeRTOS, MIPI CSI-2, I2C, SPI, UART, CAN, Oscilloscopes

EDUCATION

New York University, New York, NY — M.S., Computer Engineering

Aug. 2024 – May 2026

Blekinge Institute of Technology, Sweden — B.S., Electrical Engineering (Exchange)

Aug. 2022 – May 2023

JNTUA College of Engineering, India — B.Tech., Electronics & Communication Engineering Aug. 2019 – Aug. 2022